

Master MVA - Cours de remise à niveau

Cours n° 4 - Statistics

I - Statistical models:

definition: a statistical model is $(\Omega, \mathcal{F}, \Theta, Z)$ where

- Ω : universe, $\mathcal{F}$: sigma-algebra
- $\Theta = \{\theta \in \Theta\}$ s.t. $\forall \theta, P_\theta$ is a probability
- $Z: (\Omega, \mathcal{F}) \rightarrow (Z, \mathcal{A})$ a random variable

Model: there exists $\theta^* \in \Theta$ s.t. $Z \sim P_{\theta^*}$

Typically: $Z = (X_1, \dots, X_n)$
 $X_i \in \mathbb{R}^p$ iid

If $\Theta \subset \mathbb{R}^q$ the model is **parametric**

Example:

- Uniform $\{U([a, b]) \mid (a, b) \in \mathbb{R}^2, a < b\}$
- Gaussian $\{N(\mu, \sigma^2) \mid \mu \in \mathbb{R}, \sigma^2 > 0\}$
- Non-parametric $\{f: f \geq 0, \int f = 1\}$

definition: a model is **identifiable**
if $\forall \theta, \theta', P_\theta = P_{\theta'} \Rightarrow \theta = \theta'$

definition: a **statistic** is a random variable $T: (Z, \mathcal{A}) \rightarrow (\mathcal{T}, \mathcal{B})$
that do not depend on the true value of θ .
(θ is the unknown population parameter,
ex: $\mu, \sigma, \dots$ the mean $(\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i)$ is a statistic then)

Discussion theory

definition: a **decision rule** is a deterministic function
 $\delta: Z \rightarrow \mathcal{D}$ (decision space)

Example: • Estimation: $\mathcal{D} = \Theta$

• Binary classification: $\mathcal{D} = \{0, 1\}$

• Hypothesis testing

• Prediction/regression $\mathcal{D}: \{f: \mathcal{X} \rightarrow \mathcal{Y}\}$

definition: a loss function is $L: (\Theta, \mathcal{S}) \mapsto L(\theta, s) \in \mathbb{R}^+$

Risk: $R(\theta, s) = E_{\theta} [L(\theta, s(z))]$ → expectation for P_{θ} assuming $z \sim P_{\theta}$

(it is the abstract quantity that we want to minimize)

Example: • Estimation quadratic loss:

$$L(\theta, \hat{\theta}) = \|\theta - \hat{\theta}\|_2^2$$

$$R(\theta, \hat{\theta}) = E_{\theta} [\|\theta - \hat{\theta}\|_2^2] \text{ quadratic error}$$

Link with statistical learning: $(x, y) \sim (X, Y), (x, y) \sim \mathbb{P}$ prediction

A predictor $f: \mathcal{X} \rightarrow \mathcal{Y}$ is evaluated by the pointwise loss $l(f(x), y)$ (how bad the prediction $f(x)$ is when the true value is y)

Pointwise risk: $E_{\mathbb{P}} [l(f(X), Y)]$ (average loss over the true data distribution) $f \mapsto \hat{f}_S$ (f = predictor)

Empirical (population) risk: $\frac{1}{n} \sum_{i=1}^n l(f(X_i), Y_i)$ (average loss on the derived training sample) $S = \{(X_i, Y_i)\}_i$

Example: Prediction, squared loss:

$$l(f(x), y) = \|f(x) - y\|^2$$

The optimal prediction is $E(Y | X=x) = f(x)$

Classification (K class), log loss $-\log(q_k^{(x)})$
 $\forall 1 \leq k \leq K$ optimal $\hat{p}(Y=k | X=x)$

II - Estimation

1) Non-parametric

definition: Let $Z = (X_1, \dots, X_n)$. (X_i) iid in $(\mathcal{X}, d)$

Empirical distribution: $\hat{P}_n(z) = \frac{1}{n} \sum_{i=1}^n \delta_{X_i}$

where δ_x is the Dirac distribution

$$\forall A \in \mathcal{A}, \hat{P}_n(A) = \frac{1}{n} \sum_{i=1}^n 1_A(X_i)$$

definition: empirical moment of order k is $E_{\hat{P}}(Z^k)$

Example: empirical mean $\frac{1}{n} \sum_{i=1}^n X_i$

definition: Empirical CDF (cumulative distribution function)

$$\hat{F}_n(x) = \hat{P}_n(-\infty, +\infty] = \frac{1}{n} \sum_{i=1}^n 1_{\{X_i \leq x\}}$$

theorem (Glivenko-Cantelli): $X_1, \dots, X_n$ iid with $F_{X_i} = F$

$$\sup | \hat{F}_n(x) - F(x) | \xrightarrow[n \rightarrow +\infty]{a.s.} 0$$

theorem (Donsker, 1D): If $(X_i)_{i \geq 1}$ iid with CDF F

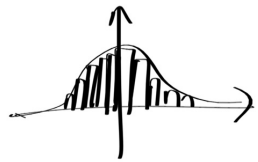
Important 

$\sqrt{n} (\hat{F}_n - F)$ converges in law to a Gaussian process of mean 0 and known Cov function of F

definition: Histogram density estimator

$$\forall n \in \mathbb{N}, f_n(x) = \sum_{i=1}^L \frac{\hat{P}_n([a_{i-1}, a_i])}{a_i - a_{i-1}} 1_{[a_{i-1}, a_i]}(x)$$

we can write as a sum



Kernel method

$$a_0 < a_1 < \dots < a_L \in \mathbb{R}$$

2) Method of moments

$\Theta \subset \mathbb{R}^d$ (parametric), $X_1, \dots, X_n$ iid $\sim P_\Theta$

the idea: estimate Θ with empirical moment

$\forall k \in \{1, \dots, q\}, m_k(\Theta) = E_\Theta[X_1^k]$
 Θ is the solution of: $\forall k \in \{1, \dots, q\}, \frac{1}{n} \sum_{i=1}^n X_i^k = m_k(\Theta) \rightarrow q$ equations

Example: $P_\Theta = \mathcal{N}(\mu, \sigma^2)$ $\Theta = \mathbb{R} \times \mathbb{R}_+^*$

$$\begin{cases} \frac{1}{n} \sum_{i=1}^n X_i = \mu \\ \frac{1}{n} \sum_{i=1}^n X_i^2 = \mu^2 + \sigma^2 \end{cases}$$

$$\hat{\theta} = \left(\underbrace{\frac{1}{n} \sum_{i=1}^n X_i}_{\bar{X}}, \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \right)$$

3) Likelihood

ol: parametric stat. model where P_θ has density p_θ
 $\hookrightarrow \mathcal{P} = \{ p_\theta \text{ density } \theta \in \Theta \subset \mathbb{R}^q \}$

definition: Likelihood function

$$L: \mathcal{Z} \times \Theta \rightarrow \mathbb{R}^+$$

$$(z, \theta) \mapsto p_\theta(z)$$

If $z = (x_1, \dots, x_n)$ iid, $L(z, \theta) = \prod_{i=1}^n p_\theta(x_i)$

definition: MLE maximum likelihood estimator

$$\forall z \in \mathcal{Z} \quad \text{MLE} = \underset{\theta \in \Theta}{\operatorname{argmax}} L(z, \theta)$$

(if it exists)

Remark: log likelihood is often useful

$$\hookrightarrow \log(L(z, \theta)) = \sum_i \log(p_\theta(x_i))$$

III - Properties of estimators

1) Risk, bias, variance

$z = (X_1, \dots, X_n)$ iid $X_i \sim p_{\theta^*} \in \mathcal{P}$

The estimator is a statistic $T_n(z)$

definition: quadratic risk $R(T, \theta^*) = E_{\theta^*} [\|T - \theta^*\|^2]$

T dominates T' if $\forall \theta, R(T, \theta) < R(T', \theta)$
 $\exists \theta', R(T, \theta) < R(T', \theta')$

definition: if $E_{\theta^*} [\|T\|] < +\infty$
 the bias of T is $b_{\theta^*}(T) = E_{\theta^*}[T] - \theta^*$

properties: if $E_{\theta^*} [\|T\|^2] < +\infty$
 Let $V_{\theta^*}(T)$ be covariance matrix of T
 Then $R(T, \theta^*) = E_{\theta^*} [\|T - \theta^*\|^2]$
 $= \|b_{\theta^*}(T)\|^2 + \text{Tr}(V_{\theta^*}(T))$

demo: $\|T - \theta^*\|^2 = \|T - E_{\theta^*}[T] + b_{\theta^*}(T)\|^2$
 $= \|T - E_{\theta^*}[T]\|^2 + \|b_{\theta^*}(T)\|^2 + 2 \langle T - E_{\theta^*}[T], b_{\theta^*}(T) \rangle$
 $R(T, \theta^*) = \text{Tr}(V_{\theta^*}(T)) + \|b_{\theta^*}(T)\|^2 + 0$
 $E_{\theta^*}[T - E_{\theta^*}(T)] = 0$
 $b_{\theta^*}(T - E_{\theta^*}(T)) = 0$

definition: If T, T' have the same bias

T is more efficient than T'

if $\forall \theta, V_{\theta}(T) \leq V_{\theta}(T')$

$\exists \theta', V_{\theta'}(T) < V_{\theta'}(T')$

(where $\forall A, B \in \mathcal{Y}_m^+(\mathbb{R}), A < B \Leftrightarrow B - A \in \mathcal{Y}^+(\mathbb{R})$)
 $\forall x, \underbrace{x^T A x} < x^T B x$

Fisher information:

Θ is an open subset of $\mathbb{R}^q$
 definition: score. If $\log(L)$ is differentiable in Θ ,

$$S_{\theta}(z) = \nabla_{\theta} \log(L(z, \theta)) \in \mathbb{R}^q$$

"Regularity assumptions"

you can differentiate twice and under the $\int$.

Requisite moments are finite.

Density supports do not depend on θ

properties: under assumptions

$$E_{\theta} [S_{\theta}(Z)] = 0$$



demo: $E_{\theta} [S_{\theta}(\theta)] = \int S_{\theta}(z) p_{\theta}(z) d\lambda$

$$= \int \nabla p_{\theta} d\lambda$$

$$= \nabla \int p_{\theta} = \nabla 1 = 0$$

definition: Fisher information

Under regularity assumptions

$$I(\theta) = V_{\theta} (S_{\theta}(z)) \quad (\text{always true})$$

$$= E [S_{\theta}(z) S_{\theta}(z)^T]$$

$$= -E_{\theta} [\nabla_{\theta} (S_{\theta}(z)^T)]$$

$$\text{ie } I(\theta) = -E_{\theta} \left[\frac{\partial}{\partial \theta} \log(L_{\theta}(z, \theta)) \right]$$

$$\begin{cases} \nabla p_{\theta} = S_{\theta}(z) p_{\theta}(z) \\ \frac{\partial \log(p_{\theta}(z))}{\partial \theta} = \frac{\partial}{\partial \theta} p_{\theta}(z) \cdot \frac{1}{p_{\theta}(z)} \\ p_{\theta} \nabla \log(p_{\theta}) = \nabla p_{\theta}(z) \end{cases}$$

$$E_{\theta} [S_{\theta}(z)] = 0$$



properties: 2 models $\mathcal{P}, \mathcal{P}'$. Some $\theta \in \mathbb{R}^q$

$$\text{Let } X \sim p_{\theta} \in \mathcal{P}, Y \sim q_{\theta} \in \mathcal{P}', X \perp Y$$

$$\forall \theta, I_{(X,Y)}(\theta) = I_X(\theta) + I_Y(\theta)$$

$$\text{density } p_{\theta}(x) q_{\theta}(y)$$

properties: $Z = (X_1, \dots, X_n)$ iid

$$I_Z(\theta) = n I_{X_1}(\theta)$$

theorem: under regularity assumption

Let $g: \mathbb{R}^q \rightarrow \mathbb{R}^d$ differentiable

T an estimator of $g(\theta)$, unbiased and has finite

moments of order 1 and 2.

If $I_2(\theta)$ is positive definite

$$V_\theta(T) \geq D_g(\theta) I_2^{-1}(\theta) D_g(\theta)^T$$

$D_g(\theta)$: matrix
Jacobiennes

If $g(\theta) = \theta$

Hammer-Roo
bund

$$V_\theta(T) \geq I^{-1}(\theta)$$

If $Z = (X_1, \dots, X_n)$ iid, $V_\theta(T) \geq \frac{1}{n} I_{X_1}^{-1}(\theta)$

definition: If $b_\theta(T) = 0$ and $V_\theta(T) = I_2^{-1}(\theta)$

Then T is called efficient.

3) Sufficiency

definition: a stat T is sufficient if the distribution of $Z|T(Z)$ does not depend on θ .

Ex: $\theta = \mu$ ($P \sim \mathcal{N}(\mu, 1)$)

$T(Z) = \frac{1}{n} \sum X_i$ is sufficient

4) Asymptotic properties

definition: If $b_\theta(T_n) \rightarrow 0$

T_n asymptotically unbiased

If $T_n \xrightarrow{prob} \theta$ it is consistent

definition: T_n is asymptotically normal if $\exists \Sigma(\theta) \in \mathcal{S}'(\mathbb{R})$
s.t. $\sqrt{n}(T_n - \theta) \rightarrow \mathcal{N}(0, \Sigma(\theta))$

property: A.N $\Rightarrow$ consistent

we can compare asymptotically Σ

definition: This implies a definition of being asymptotically more efficient

definition: asymptotically efficient
 $\Sigma(\theta) = \Sigma^{-1}(\theta)$

theorem: under some hypothesis
the MLE is consistent, A.N and asymptotically efficient.